Evolutionary Computation
CA2 Report

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Exercise 4

μ, λ Binary Genetic Algorithm with Crossover for MAX-SAT Optimisation
Python Implementation: `./common/maxsat.py`

```
// Input variables
let n           = number of variables
let m           = number of clauses
let clauses     = list of clauses
let time_budget = maximum amount of time (in seconds) allowed for the algorithm to run

// Algorithm parameters
mu_count        <- 5
lambda_mu_ratio <- 6
crossover_prob  <- 0.85
mutation_prob   <- 1 / n
lambda_count    <- mu_count * lambda_mu_ratio

start <- time_s()             // store start time
end   <- start + time_budget  // pre-compute end time

// Initialise Random Population
population <- empty set
for i <- [1..mu_count]
  population[i] <- uniformly random list of integers from {0,1} of size n
end for

// Main Loop
generation <- 0
xbest      <- empty
nsat       <- -1
while time_s() <= end

  generation <- generation + 1

  // select lambda_count offspring
  offspring <- empty set
  for i <- [1..lambda_count]
    offspring[i] <- random individual from population
  end for

  // apply crossover
```

```
for i <- [1,3,5,...,lambda_count-1]  // step by 2
  if rand() < crossover_prob
    o_1, o_2         <- offspring[i], offspring[i+1]
    crossover_point  <- rand([1..n-1]) // random int between 1 and n-1
    offspring[i]     <- concat(o_1[1:crossover_point], o_2[crossover_point+1:n])
    offspring[i+1]   <- concat(o_2[1:crossover_point], o_1[crossover_point+1:n])
  end if
end for

// apply mutation
for i <- [1..lambda_count]
  for j <- [1..n]
    if rand() < mutation_prob
      offspring[i][j] <- offspring[i][j] ^ 1
    end if
  end for
end for

// calculate offspring costs
costs <- empty set
for i <- [1..lambda_count]
  individual        <- offspring[i]
  satisfied_clauses <- 0
  for each clause in clauses
    clause_satisfied <- false
    for each literal in clause
      value    <- individual[abs(literal)]
      if literal < 0
        value <- value ^ 1
      end if
      if value == 1
        clause_satisfied <- true
      end if
    end for
    if clause_satisfied
      satisfied_clauses <- satisfied_clauses + 1
    end if
  end for
  costs[i] <- satisfied_clauses
end for

// truncation selection
idxs       <- sort_desc(costs)[1:mu_count] // select mu_count best solutions
population <- offspring[idxs]

// update record of best individual
best_offspring <- offspring[idxs[1]]
best_nsat      <- costs[idxs[1]]
if best_nsat > nsat
  xbest <- best_offspring
  nsat  <- best_nsat
end if

// if we've covered every row, we can exit early.
if best_nsat == m
  break while
end if
```

```
end while

t <- generation * mu_count
return t,nsat,xbest
```

Exercise 5

Three parameters were chosen to run experiments against:

- `lambda_mu_ratio`: The ratio of $\lambda : \mu$ (where $\mu = 5$)
- `crossover_prob`: The probability of applying a single-point crossover between 2 offspring
- `mutation_prob`: The probability of flipping a single bit in the binary representation of an individual

For all three parameters, the same three benchmark instances were used:

- `bcp-fir/normalized-fir06_area_delay.wcnf`
- `reversi/rev66-4.wcnf`
- `pbo-mqc-nencdr/10tree305p.wcnf`

Experiments: `lambda_mu_ratio`

For the experiments modulating the `lambda_mu_ratio` parameter, the following values were chosen:

`lambda_mu_ratio`	2	3	4	5	6	7	8	9	10

For each parameter setting, 100 independent trials were performed with a 5 second time limit per trial. The figure reports the resulting boxplots, split by benchmark instance.

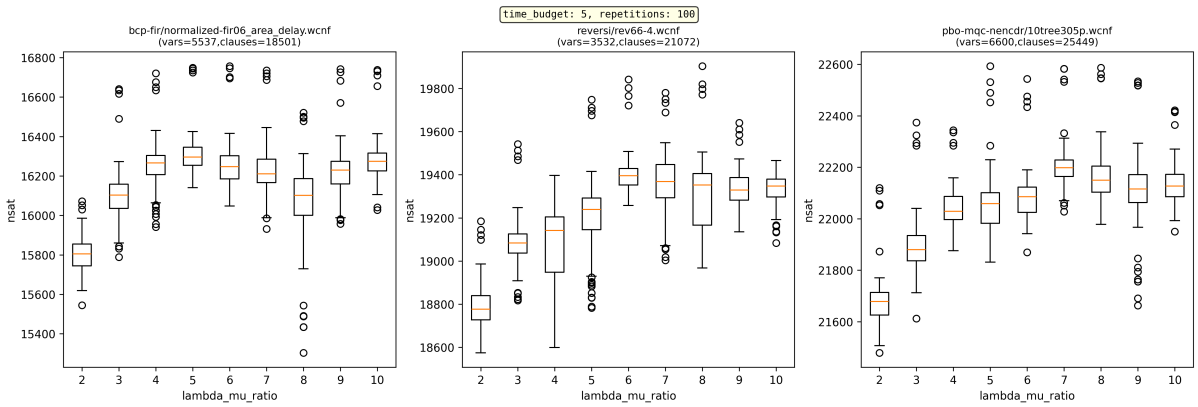

Figure 1: `lambda_mu_ratio` trial results

Discussion:
The results from the above trials show that increasing the `lambda_mu_ratio` improves solution quality up to a point, beyond which further increases yield diminishing returns. Across all three benchmark instances, performance is lowest at ratio 2 and improves notably when ratio is increased to 3, and again when increased to 4. From ratio 5 onwards, the median solution quality plateaus, with minimal improvement at higher ratios (7–10). This suggests that the additional selection pressure from larger offspring pools provides substantial benefit initially

by allowing better parents to be selected, but eventually the marginal improvement becomes negligible.

However, the plateauing effect may also be influenced by the fixed time budget. Higher `lambda_mu_ratio` values require generating and evaluating more offspring per generation, which increases computational cost per iteration. Under this time limit, higher ratios result in fewer total generations completed, offsetting the potential benefits of stronger selection pressure. This creates a trade-off where larger offspring pools appear to improve solution quality per generation, but fewer generations can be executed within the time bound. The consistent pattern across all instances indicates that a ratio of 5–6 represents a reasonable balance between computational cost and quality gain, accounting for both the selection benefit and the generational cost. Existing academic research typically suggests a value of $\sim$7 for the $\lambda : \mu$ ratio, which these experiments support within a margin of problem and implementation-specific variation.

Experiments: `crossover_prob`

For the experiments modulating the `crossover_prob` parameter, the following values were chosen:

`crossover_prob`	0	0.1	0.5	0.75	0.8	0.85	0.9	0.95	1

For each parameter setting, 100 independent trials were performed with a 5 second time limit per trial. The figure reports the resulting boxplots, split by benchmark instance.

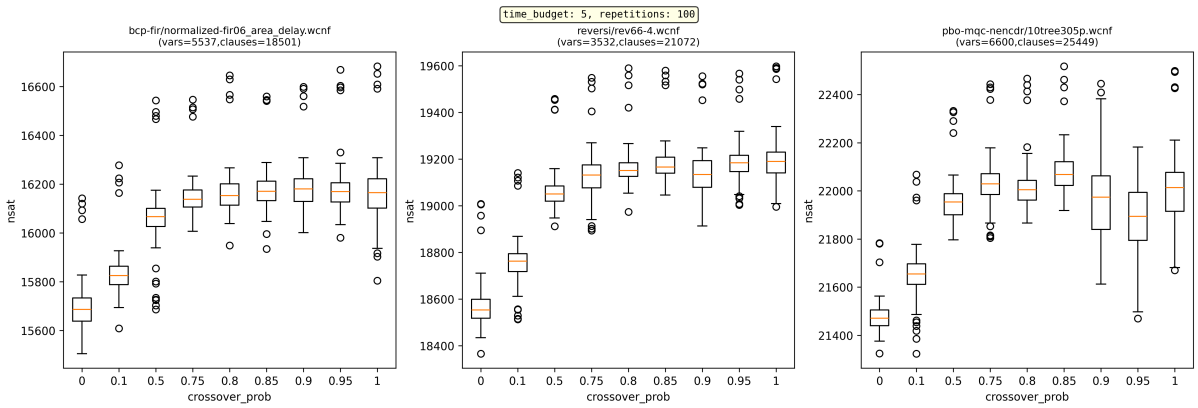

Figure 2: `crossover_prob` trial results

Discussion:
The results from the above trials show that crossover probability has a clear impact on solution quality, with performance varying substantially across the tested range. At `crossover_prob` of 0 (no crossover), all three instances exhibit notably poor solutions, indicating that crossover is beneficial for this problem domain. Performance improves steadily as crossover probability increases from 0, reaching a peak in the range of 0.75–0.85 where median solution quality is best. Beyond 0.75–0.85, increasing the crossover probability further shows either stagnation or slight degradation on some instances.

The base algorithm choice of `crossover_prob` = 0.85 aligns well with these experimental results and represents an effective configuration across diverse problem instances.

Experiments: `mutation_prob`

For the experiments modulating the `mutation_prob` parameter, the following values were chosen:

`mutation_prob`	1/n	5/n	10/n	15/n	20/n	25/n

For each parameter setting, 100 independent trials were performed with a 5 second time limit per trial. The figure reports the resulting boxplots, split by benchmark instance.

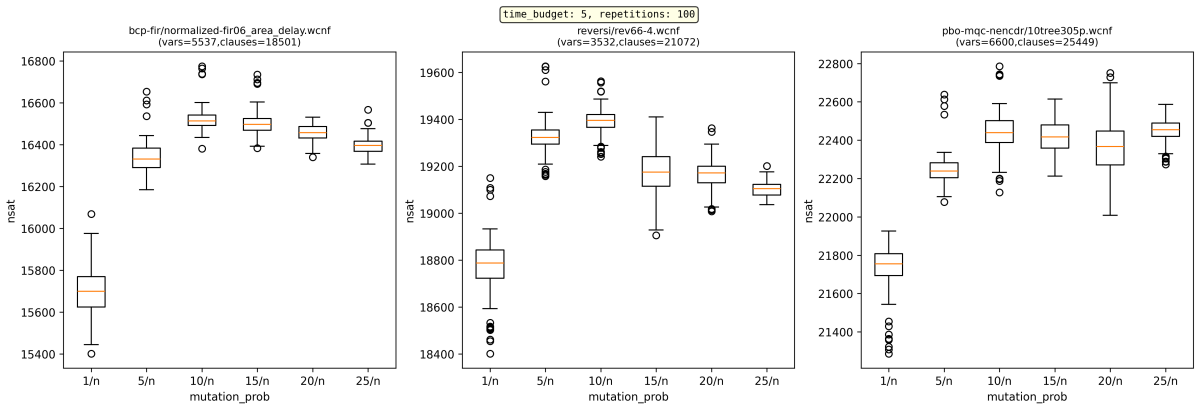

Figure 3: `mutation_prob` trial results

Discussion:
The results from the above trials show an unexpected pattern: high mutation probabilities (e.g. 10/n, 15/n, and 20/n) yield better solution quality compared to lower mutation rates (e.g. 1/n, 5/n). This result contradicts conventional wisdom in evolutionary computation, where mutation rate is typically suggested to be set to 1/n. I believe that these results are due to the short 5 second time limit. Within a short time window, higher mutation rates may provide exploration benefits that outweigh their disruptive effects, helping the algorithm escape local optima and sample the search space more thoroughly before time expires. Conversely, lower mutation rates on short time budgets may become too exploitative prematurely, converging to mediocre solutions without sufficient time to refine them.

To confirm this hypothesis, additional experiments were conducted with a longer 20 second time budget on the `bcp-fir/normalized-fir06_area_delay.wcnf` instance (Figure 4). The results strongly support the time budget theory: with adequate time, lower mutation rates (1/n and 5/n) significantly outperform higher rates. In contrast, higher mutation rates (15/n, 20/n, 25/n) plateau at lower quality, suggesting they converge to suboptimal solutions faster.

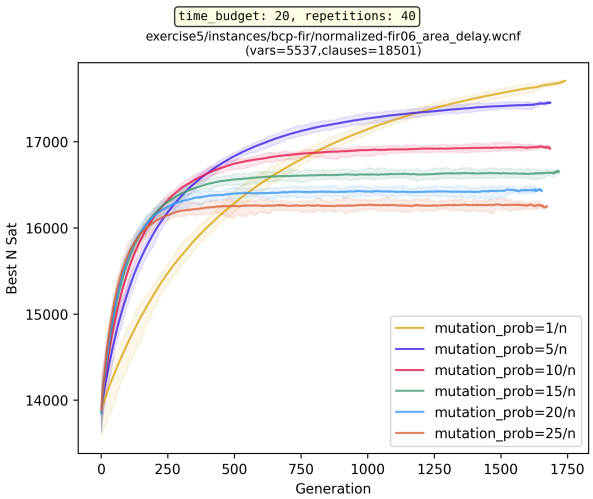

Figure 4: `mutation_prob` convergence over time (20s budget, single instance)

The explanation is clear: with only 5 seconds, the algorithm explores broadly with high mutation but lacks time to refine solutions, making the initial diversity beneficial. With 20 seconds, the extended time allows lower mutation rates to perform refining exploitation, yielding

superior final solutions. These findings demonstrate that mutation rate efficacy is highly time-budget dependent. For the 5-second constraint used in the main experiments, a high mutation rate (10/n–15/n) is effective; however, for longer runs with adequate convergence time, the base algorithm's conservative choice of 1/n is indeed optimal.

Given the clear time-budget dependency observed, an adaptive mutation strategy that decreases the rate over time could potentially combine the benefits of both regimes. Such an approach would provide high initial population diversity to escape local optima and sample broadly, while transitioning toward lower mutation rates as the time budget approaches exhaustion, allowing exploitation and refinement of promising solutions. This strategy shares conceptual similarity with simulated annealing, which gradually reduces temperature (exploration intensity) over time to balance exploration and exploitation. Such an approach could represent a promising direction for improving this algorithm's performance across time budgets and problem instances.